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Development of a single degree of freedom perforate impedance model under grazing flow and high SPL

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This paper reports on both no flow normal impedance tube tests, and grazing flow in-situ impedance tests, performed on single layer perforate liner configurations, typical of aircraft engine applications, at high sound pressure levels (up to 155dB), and at high grazing flow Mach numbers (up to M0.7). No flow results show the dominant non-linear resistance component, along with a reduction in non-linear acoustic resistance with increasing frequency. This reduction arisies from an increase in facing sheet hole discharge coefficient, in line with Zhang and Bodony's numerical studies, which show that the hole mass flux is closely linked to the incident sound frequency. Grazing flow results show that the dominant resistance contributions arise from a linear grazing flow resistance component, which also reduces with increasing frequency, along with a significant non-linear resistance component, which is largest for low open areas, and varies with acoustic velocity, and hence also SPL and frequency. Semi-empirical models for the impedance under grazing flow are developed using fits to the measured data, following on from the approaches of Kooi and Sarin, and Rice, with the addition of a non-linear resistance component. Comparisons of prediction with measurement for both pure tone and broadband sources show excellent agreement for single layer designs typically employed in aircraft engines. These models highlight the importance of the non-linear resistance component, particularly for designs placed close to the engine fan.

Nomenclature

POA	=	percentage open area
P	=	acoustic pressure
SPL	=	sound pressure level
SDOF	=	single degree of freedom
OASPL	=	overall sound pressure level
EPNL	=	effective perceived noise level
Vi	=	incident acoustic velocity
M	=	Mach number
V*	=	skin friction velocity
V_e	=	flow velocity at edge of boundary layer
τ	=	shear stress
c_f	=	skin friction coefficient
R	=	acoustic resistance
X	=	acoustic reactance
FDF	=	(NLR) flow duct facility
K	=	wave number
ρ	=	air density
c	=	speed of sound
f	=	frequency
m_f	=	facing sheet inertance

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t	=	perforate sheet thickness
ε	=	perforate sheet thickness end correction
d	=	perforate sheet hole diameter
C_D	=	hole discharge coefficient
L	=	cell height

I. Introduction

Noise has become a key design driver for future aircraft designs. This is primarily due to the proliferation in the number of noise events in the vicinity of airports, and the subsequent increase in sensitivity of the local population to aircraft noise. It is therefore imperative to ensure that all of the components which affect the overall noise levels on the ground (engine, nacelle, airframe, flight operations) are fully optimised to exploit their potential noise reduction contributions.

Aircraft engine ducts are typically lined wherever possible with acoustic panels, for the reduction of radiated noise. It follows that, in line with progress experienced in engine source noise reduction, jet noise and airframe noise reductions, it is essential that aircraft engine duct acoustic panels are designed to present an impedance to the propagating noise which is as closely matched as possible to the optimum impedance desired for maximum attenuation, for each of the flight certification points, at the propagation angles which contribute most to EPNL. The acoustic panels have one or more resistive facing sheets, separated by one or more honeycomb layers, with the overall panel being backed by a highly reflective backing sheet. The resistive sheets may be either perforated materials or resistive meshes.

Single layer perforate liners are the simplest constructions to manufacture, and hence have the lowest cost. They are also lightweight and robust, making them widely used in aircraft engines. One complication in their use is that these liners show a strong acoustic resistance sensitivity to grazing flow, which is much less pronounced in other more expensive constructions, such as single degree-of-freedom liners with wire mesh facing sheets, or double degree-of-freedom liners, where the septum resistive layer provides the majority of the total resistive component.

This study seeks to improve the understanding of the acoustic characterisation of single layer perforate panels, particularly their grazing flow impact and the non-linear resistance component under grazing flow. It describes tests to aid the development of a semi-empirical single degree of freedom model which predicts the installed impedance as a function of each of the panel geometric parameters, and the installed flow and incident noise source conditions.

II. Background

Over the last 40 years, many semi-empirical perforate liner acoustic impedance prediction models[1-11] have been developed. A review of some of the existing single layer perforate models in open literature shows that large differences in predicted impedance, and in particular, acoustic resistance, may exist for a fixed construction. Figure 1 shows predictions from some of the most commonly used industry models for a relatively low open area inlet SDOF perforate @ M0.3, akin to an Approach condition. Differences in resistance of almost 2ρc are seen at low frequencies. It is also noted that the majority of the predictions are (almost) constant with frequency, and as liners are typically sized for EPNL-influencing higher frequencies (e.g. ~1KHz to 6KHz), the respective differences become greater, and even more important.

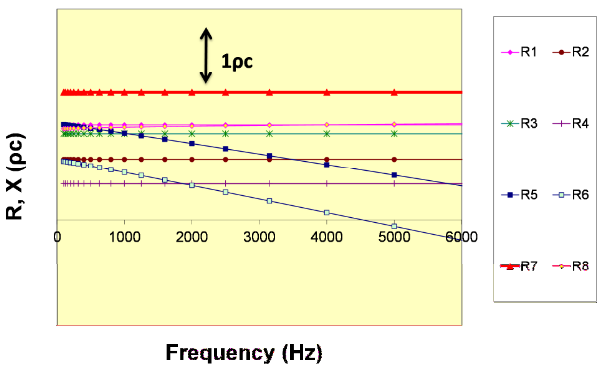

Figure 1. Single Layer Perforate Impedance Model Resistance Comparison, M0.3

The models used in Fig 1 have been derived largely from fits to insertion loss data, with some arising from fits to in-situ impedance measurements. While manufacturing variability must be taken into account[11] (hole blockage, differing hole shapes from varying perforation procedures and material types), the large predicted impedance variations emphasise the need for improved prediction models for these low cost, lightweight, but acoustically complex designs, opening up the potential for improved attenuation from the refinement of the geometry of existing and future applications.

Varying approaches have been used to depict the grazing flow impedance effect. Most of the existing perforate resistance models relate the acoustic resistance to the grazing flow Mach number, M. Kooi and Sarin[4] sought to investigate the influence of the variation in boundary layer characteristics for a constant mean duct Mach number, arguing that the skin friction velocity characteristic of the inner boundary layer is a more appropriate correlation parameter, when the boundary layer thickness is significantly higher than the perforate hole diameter. This is the situation experienced both in test ducts and in engine ducts. Artificially thickening the test duct boundary layer at constant Mach number, Kooi and Sarin produced an excellent correlation between the acoustic resistance and the skin friction velocity parameter. A similar modelling approach was also followed by Malmary et al[7].

The work reported in this paper reports on recent in-situ grazing flow impedance tests carried out in the NLR grazing flow duct facility in Holland, which have taken advantage of new, more powerful, speakers. The higher SPLs generated by these speakers have allowed the measurement of greater non-linear increases in installed perforate liner impedance. This paper, following on from the earlier work of Ref 4 and Ref 11, documents the development of an improved SDOF perforate impedance model under grazing flow and high amplitude sound sources. The work shows that the installed engine acoustic resistance has a significant non-linear component at high SPLs for the low POAs required for aircraft engine inlet applications, and which remains important for the design of the relatively higher open area configurations used for bypass duct applications.

III. Perforate Liner Test Configurations

Three single layer perforate configurations were chosen for study. Low, medium and high facing sheet geometric Percentage Open Areas (POAs) were chosen, to cover the range of interest applicable for lined single layer inlet (low POA), and bypass (medium, high POA), aircraft engine duct applications. NLR manufactured a test panel with three regions of varying open area, and a square hole pattern. The facing sheet was mechanically drilled, and was backed by 3/8" width, Al hexagonal core, and a rigid Al backing sheet. The panel depth was chosen to ensure the resonance for each perforated zone lay in the region of maximum accuracy for 29mm impedance tube (1KHz to 5KHz) tests, and the NLR grazing flow tests. For grazing flow in-situ impedance testing, 3 cells were selected for instrumentation, one from each open area zone. The local geometric POA was calculated for each of the instrumented cells.

IV. Normal Incidence Acoustic Impedance Measurements

The fully bonded acoustic panel was impedance tested, first without, and then with, the presence of grazing flow. Initial pure tone and broadband normal impedance measurements were made to evaluate the fundamental no flow characteristics of the SDOF perforate configurations.

Normal incidence acoustic impedance measurements were performed on the three configurations using the Brüel & Kjær high intensity portable impedance meter[19]. The Brüel & Kjær hand-held flanged kundt tube has been developed for the aerospace industry, for both production acoustic quality control, and research, impedance measurements, with a capability to test at SPLs up to 155dB. Dedicated routines allow the measurement of key aircraft acoustic panel quality control parameters, such as impedance spectra versus OASPL, and acoustic resistance versus acoustic velocity, for either a broadband or a pure tone source.

Fig 2 shows an example of the measured broadband noise impedance spectra, for the medium POA liner configuration, at OASPLs between 130dB and 150dB, increasing in 5dB steps. As expected, as OASPL increases, the resistance increases, and the resonance frequency increases.

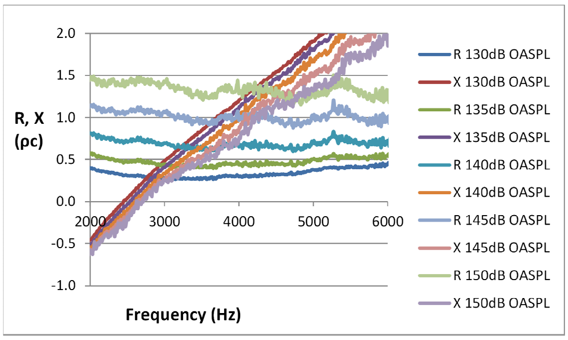

Figure 2. Medium POA Normal incidence impedance tube OASPL spectra

The no grazing flow resonance frequency reduced with decreasing POA, consistent with the increase in facing sheet mass reactance with reducing open area. As open area reduced, the liners also showed the expected relative increases in non-linear resistance with increasing OASPL.

The portable impedance meter allows the source to loop with either broadband or pure tone signal. Ref 11 showed that the traditional DC flow resistance quality control characteristic for SDOF perforates could be replicated with acoustic measurements using an increasing pure tone level at the panel resonance frequency. The resistance characteristic approximates the form of a straight line plus intercept. The intercept arises from viscous scrubbing friction in the holes, and is relatively small. The slope defines the primary acoustic loss mechanism under normal incidence and zero grazing flow at high SPLs. Refs. 12 and 16 showed that this acoustic loss occurs via the conversion of acoustic kinetic energy into viscous losses, via the creation of aerodynamic vortices at the hole entry and exit. The measured slope may be related directly to the open area, POAeff, as follows[8]:-

$$NoflowSlope = \frac{1}{2}\frac{\rho}{\left(C_D POA_{eff}\right)^2} \qquad (1)$$

where ρ is the density, while C_D is traditionally assumed = 0.76[8]. If a consistent set of assumptions are assumed (e.g. C_D = 0.76, DC = acoustic POAeff at resonance frequency), effective POA can be used to cross-correlate different measurements from different facilities.

Further to the computational modeling work of Zhang and Bodony[18], where the mass flux through an aperture was shown to vary with frequency, non-linear acoustic impedance measurements were performed on the three configurations reported here, at a range of pure tone frequencies between 2000Hz and 5000Hz, for SPLs between 130dB and 150dB. At each frequency/ SPL, the impedance meter calculates the acoustic resistance and the associated acoustic velocity (= P/ |Z|).

For each configuration, and for each pure tone test frequency, the resistance/ velocity characteristic was approximated by a straight line fit. Figure 3 presents the calculated discharge coefficients. For a given geometric open area, as frequency increases, the slope tends to reduce, reflecting lower non-linear losses for a given incident acoustic velocity. This behaviour reflects a general increase in time-averaged discharge coefficient with increasing frequency, and an equivalent reduction in time-averaged displacement thickness with increasing frequency[18].

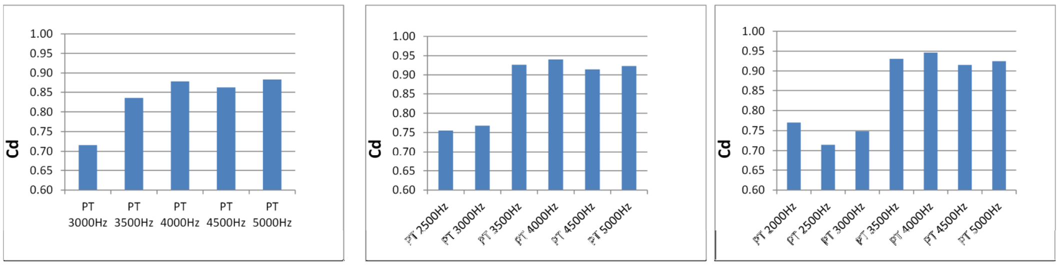

Figure 3. Extracted C_D from no grazing flow normal incidence acoustic resistance-acoustic velocity data (left to right:- high, medium and low POA configurations)

It will be seen later that the measured grazing flow resistance component also reduces with increasing frequency.

V. NLR Test Facility Description and Test Procedure

The three SDOF perforate configurations were in-situ impedance tested under grazing flow in the NLR Aerospace Acoustic Laboratory in Holland. The samples were instrumented with microphones for in-situ measurements, with and without the presence of grazing flow, in their Flow Duct Facility (FDF).

Figure 4 shows a schematic of the NLR FDF. The facility is a suction tunnel that consists of two reverberation rooms connected by a duct with a test section length of 1.05m. The section dimensions are 0.15m x 0.30m. The maximum duct Mach number is 0.7. The sample sizes for in-situ measurements are typically 0.17m (0.01m lip each side) x 0.17m. Insertion loss measurements can also be performed, with sample sizes of 0.17m x 0.85m.

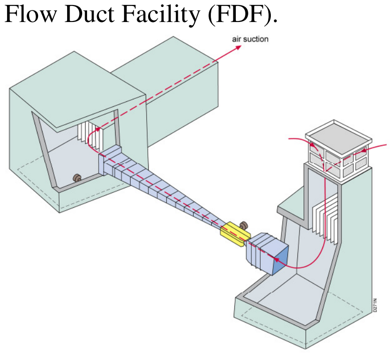

Figure 4. NLR Flow Duct Facility (FDF)

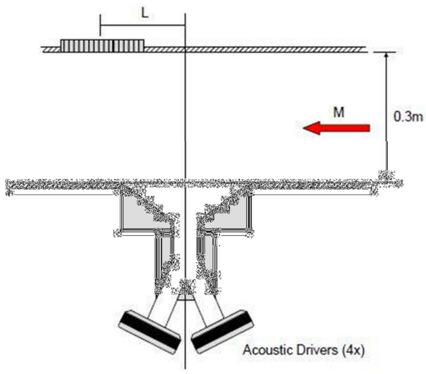

Figure 5. NLR in-situ impedance measurement set-up

Figure 5 shows a schematic of the test set-up for in-situ measurements. The sound source was both broadband and tonal. The broad band source was generated in the upstream reverberation room, at an overall SPL of up to 145dB. The pure tone source was generated using a signal generator, which produced tones from 4 loudspeakers, at frequencies from 500Hz to 6000Hz, with a step size of 500Hz. The maximum pure tone level has recently been increased with the introduction of new speakers, allowing the realization of SPLs up to 155dB. Two sets of individual pure tone levels were tested, 130dB, and the maximum achievable.

Miniature (1.65mm diameter) Endevco microphones were flush mounted and sealed in the sample facing sheet and backing sheet, at three cell locations per sample. The impedance was calculated using the cross power spectrum via the following equation[20],

$$Z = \frac{P_1^* P_0}{P_1^* P_1} \frac{-i}{\sin KL} \qquad (2)$$

where P_0 is the complex acoustic pressure measured at the sample facing sheet, P_1 is the complex acoustic pressure measured at the sample backing sheet, * denotes the complex conjugate, $i = (-1)^{1/2}$, K is the wave number ($=2*\pi*f/c$) and L is the cell height. As the measurement is inherently local, it is important to account for any variations in geometric POA between the instrumented cells when reducing the test results. All of the test data was corrected for the relatively small blockage effects of the microphones, using correction equations provided by NLR.

VI. The influence of grazing flow

It is well known that grazing flow significantly affects the resistance of perforate panels. Recent studies[12-18] have provided more insight regarding the mechanism for acoustic absorption under the presence of grazing flow. Dupère[16] stated that in the presence of mean flow, the acoustical shed vorticity at the upstream edge of the Helmholtz resonator neck opening is swept away by the presence of the flow. He also pointed out that the incident acoustic energy is transferred from the acoustic waves into the vortical part of the flow field, where it is eventually dissipated as heat. Furthermore, Dupère's studies suggested that that there was a fundamental difference between the cases without and with flow, adding that without flow, the absorption coefficient is a non-linear function of incident acoustic pressure amplitude, while in the presence of grazing flow the absorption coefficient is independent of the incident acoustic pressure amplitude. Murray et al's[11] grazing flow measurements showed that indeed the acoustic losses remained high for SDOF perforates under grazing flow and relatively low SPLs, while for the relatively moderate SPLs levels achievable in the test, there was only a small increment in acoustic resistance.

With reference to the influence of the boundary layer characteristic, Kooi and Sarin[4] investigated the effect on the grazing flow impedance. Their work was performed in the NLR tunnel, where, by artificially thickening the tunnel boundary layer (through normal blowing), they showed that the resistance was closely correlated to the measured skin friction velocity, rather than the tunnel Mach number. The data showed a reduction in perforate panel resistance with reducing skin friction velocity, for a fixed tunnel centreline Mach number. Kooi and Sarin went on to develop a correlation for the resistance based on skin friction velocity. The skin friction velocity provides a representation of the velocity in the vicinity of the perforate holes and accounts for the shape of the boundary layer. The skin friction velocity, V*, is defined as,

$$V^* = \left(\frac{\tau_w}{\rho_w}\right)^{\frac{1}{2}} = \left(\frac{c_f}{2}\right)^{\frac{1}{2}} * V_e \qquad (3)$$

where τ_w is the local shear stress at the wall, ρ_w is the local density, c_f is the skin friction coefficient, and V_e is the local velocity at the outer edge of the boundary layer. This approach was also adopted by Malmary[7]. Zandbergen et al[5] reported on the first measurements of in-flight resistance for a Fokker F28 inlet, where they confirmed the correlation between resistance and skin friction velocity.

Recent CAA modelling studies by Zhang and Bodony[18] indicate that the perforate facing sheet hole time varying displacement thickness, and hence hole discharge coefficient, varies with frequency, driving a variation in acoustic resistance with frequency. The authors also showed that, at the relatively low Mach number of M0.15, there was a clear increase in acoustic resistance when the incident SPL was increased from 130dB to 150dB.

The NLR tests reported in this paper were aimed at measuring the installed in-situ impedance under grazing flow conditions for the NLR manufactured samples, under both moderate (~130dB), and high (>150dB), SPLs. The initial test goal was first to measure the variation in acoustic resistance with grazing flow Mach number (whose relationship with skin friction velocity is a constant factor for a straight test duct), with SPL, and with frequency. Finally, a semi-empirical model would be developed to fit to the measured data.

VII. NLR In-situ impedance measurements

In-situ impedance measurements were performed on the three SDOF perforate configurations at Mach numbers of 0, 0.3, 0.5, and 0.7. Ref 11 has described the excellent level of agreement between the NLR in-situ measurements at M = 0, and normal incidence impedance tube measurements, hence ensuring there is good read-across between the no flow impedance tube data, and the grazing flow in-situ data. One cell in each of the three POA sections of the sample was selected for instrumentation. Pure tone tests were performed at 130dB, and the maximum achievable SPL, at frequencies between 500Hz (max achievable ~155dB) to 6000Hz (max ~140dB), in 500Hz steps, while broadband tests were performed at 130dB OASPL, and the maximum achievable (~145dB OASPL). At low Mach numbers/ SPLs, tests were also performed at 160Hz and 240Hz.

Figure 6 shows the pure tone no flow NLR in-situ impedance for the medium POA configuration. There is a clear non-linear resistance response, which peaks around the cell resonance frequency, and which varies with the varying pure tone SPL at each frequency. In addition to the non-linear resistance variation, there is also evidence of a non-linear reactance variation with SPL, as the reactance curve deviates slightly from a smooth trend, e.g. at 3000Hz, for the configuration shown, where the SPL is highest. This is a reflection of the facing sheet reactance component reducing due to a reduction in the hole end correction at increased SPL.

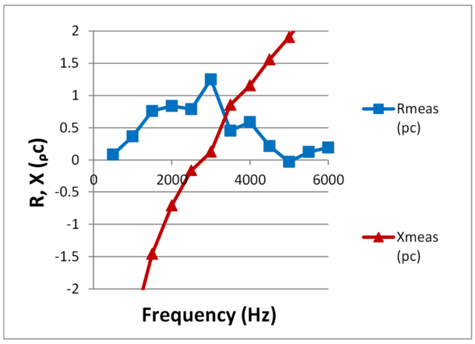

Figure 6. In-situ impedance, medium POA SDOF, M = 0, maximum Pure Tone SPL

Figure 7 shows an example of the pure tone data, for the medium POA liner with a grazing flow centerline Mach number of 0.3, for the maximum achievable SPL.

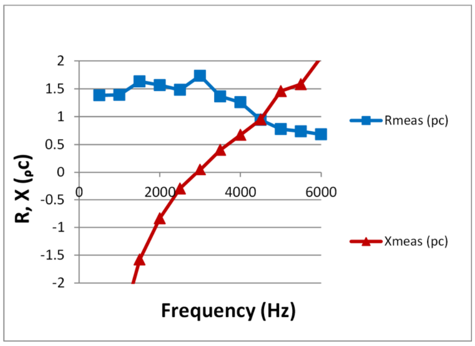

Figure 7. In-situ impedance, medium POA SDOF, M = 0.3, maximum Pure Tone SPL

Comparison of grazing flow data with no flow data shows clear variations due to the addition of the grazing flow. The acoustic resistance increases for all configurations, with the largest increase seen at low frequencies. This is in line with Zhang and Bodony's[18] grazing flow modeling studies, which indicated that incident waves of lower frequency lead to stronger vortex shedding. Hence, the grazing flow is adding a resistance contribution which varies with frequency.

Figure 8 shows the measured impedance at M0.3, 0.5, and 0.7, for the low POA configuration, at maximum achievable SPL. It is seen that that the acoustic resistance increases with Mach number, and continues to fall-off with increasing frequency (though the rate of fall-off is SPL, as well as frequency, dependent). The resonance frequency also increases with increasing Mach number, reflecting reductions in the facing sheet reactance component. Finally, as Mach number, and acoustic resistance increases, there is a reduced non-linear increment for a given tone SPL.

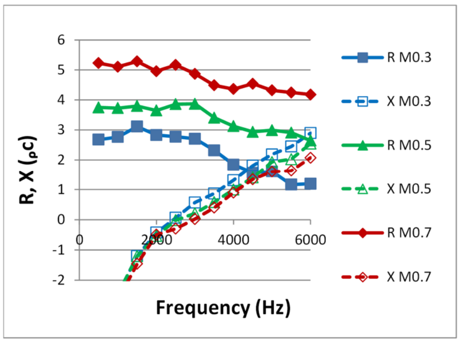

Figure 8. In-situ impedance, low POA SDOF, M = 0.3, 0.5, 0.7, maxdB Pure Tone SPL

As stated earlier, in the NLR tunnel, which is designed to have no pressure gradient, there is a direct relationship between the centerline Mach number and the skin friction velocity. A first order fit to the resistance was attempted, based on the use of Mach number as a correlation parameter. Given the direct relationship between the tunnel centerline Mach number and skin friction velocity, this correlation can subsequently be converted to a correlation with the latter. In a nacelle inlet, both favourable (lip contraction), and adverse (diffusion along the inner barrel), pressure gradients exist, which alter this relationship. To this may be added the flight effects of nacelle incidence and aircraft forward speed. The incidence effect increases variations in the boundary layer around the inlet, while the forward speed effect leads to shorter boundary layer wrap distances from the stagnation point for flight than for ground tests. Pressure gradients are also seen in the bypass duct, further emphasizing the need for correlations based on boundary layer characteristics.

The resistance equations follow the basic form developed by Kooi and Sarin[4], with altered multiplying constants, and with the addition of the non-linear term, while the reactance terms follow the form of the equations reported by Rice[1], with a reduced dependence on Mach number. The resulting normalised resistance fit has the following form:-

$$R = R_{vis} + R_{gf} + R_{nl} \qquad (4)$$

where R_{vis} represents the viscous hole losses, R_{gf} the linear grazing flow losses, and R_{nl} the non-linear losses, and,

$$R_{vis} = \frac{k_1 \mu t}{\rho c \sigma C_D d^2} \quad (5), \quad R_{gf} = \frac{k_2 M\left[5 - \left(\frac{t}{d}\right)\right]}{4\sigma} - \frac{k_3 d f}{\sigma c} \quad (6), \quad R_{nl} = u * slope = \frac{k_4 * u * (1-\sigma^2)}{2c(C_D \sigma)^2} \quad (7)$$

and k_i, i=1 to 4, are constants. The acoustic velocity, u, is calculated iteratively from P/|Z|,

$$u(cm/s) = P(dynes/cm^2) / \left\{\left[\left(R_{vis} + R_{gf} + u * slope\right)\rho c\right]^2 + [(X_m + X_c)\rho c]^2\right\}^{0.5} \qquad (8)$$

with $\rho(g/cm^3)$, and $c(cm/s)$, in consistent cgs units. The normalised reactance is given by,

$$X = X_m + X_c \qquad (9)$$

where,

$$X_m = k(t + \varepsilon d)/\sigma,\ \varepsilon = \frac{0.85(1-0.7\sigma^{0.5})}{(1+200M^3)},\ \text{and}\ X_c = -\cot(kL) \quad (10)$$

Figure 9 shows a comparison between the prediction and measurement at M0.3 for the low POA design. The model shows excellent agreement with the measurements. The grazing flow component falls off linearly with increasing frequency (see eqn 6). It is also noted that, although the grazing flow resistance component is the dominant contributor in all cases, there remains a significant non-linear component at high SPLs, which is tracked by the model.

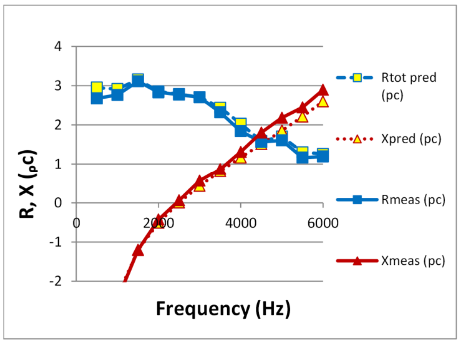

Figure 9. Low POA SDOF, in-situ impedance, prediction vs measurement, M = 0.3, maxdB Pure Tone SPL

Test results for the other POAs and Mach numbers also show excellent agreement with prediction, confirming that the semi-empirical approximation holds for all cases.

It is noted, that for engine liners, particularly those placed close to the engine fan, the non-linear component will increase further at high power settings, given SPLs may be significantly higher than those seen in this test. This further emphasizes the need for accurate non-linear resistance modeling.

Figure 10 shows prediction versus measurement for the low POA configuration, at M0.5, for the maximum achievable broadband signal. The comparisons confirm the prediction capability remains good for broadband noise.

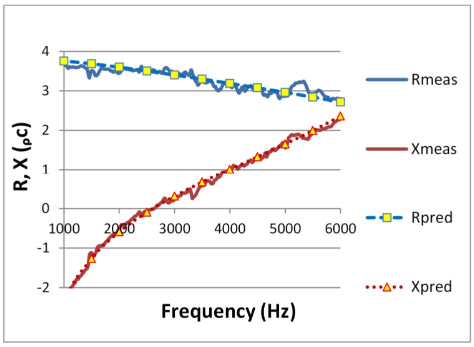

Figure 10. In-situ impedance, prediction vs measurement, low POA SDOF, M = 0.5, maxdB Broadband SPL

VIII. Conclusion

This paper has reported on investigations into the effects of grazing flow and high sound pressure levels on the acoustic response of signal layer perforate panels, of types representative of those used in aircraft engine inlet and bypass ducts. Normal incidence impedance tube measurements were made to evaluate the non-linear acoustic response under zero grazing flow conditions, while in-situ impedance measurements were then performed in the NLR grazing flow facility to evaluate the change in impedance characteristic when grazing flow is added to high incident sound pressure levels.

The zero grazing flow impedance tube measurements for increasing pure tone levels at varying frequency showed how the non-linear acoustic resistance slope reduces as frequency is increased, reflecting a reduction in the time-averaged hole boundary layer displacement thickness, and a corresponding increase in discharge coefficient. The grazing flow results showed that the presence of the flow significantly alters the acoustic impedance characteristic, with significant low frequency resistance being generated at frequencies below the resonance. The grazing flow resistance component was shown to reduce with increasing frequency, suggesting the influence of a reducing hole boundary layer thickness as frequency increases, also under grazing flow. The in-situ data also highlighted that significant non-linear resistance components continue to contribute to the total liner resistance, and must be accounted for when sizing aircraft engine duct acoustic panels. The non-linear component is greatest for low open areas and high sound pressure levels.

Semi-empirical impedance models were adapted to the measured impedance trends with effective open area and duct centerline Mach number, following the approach of Kooi and Sarin[4,] and Rice[1], and employing a non-linear resistance prediction which varies with acoustic velocity (and hence SPL and frequency). The impedance prediction was shown to provide excellent agreement with the measured data for a range of open areas covering traditional nacelle applications. Finally, predictions under broadband conditions also showed good agreeement with the measured data. The new models can now be employed to provide better-sized, and hence more efficient, configurations for engine inlet and bypass duct liner applications, particularly for those liners placed close to the fan.

IX. Acknowledgements

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